Martin McEnroe, P.E.

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ENTERPRISE AGENTIC AI ARCHITECT

Two U.S. Patents Filed 2026—AI Agent Safety • 22 Patents Issued • Licensed P.E. • IEEE PES SC-5 Co-Chair

Architects and governs enterprise agentic AI systems, with three production platforms in service: multi-agent orchestration with cross-vendor adversarial verification, autonomous conversational agents with layered evaluation, and consumer AI with defense-in-depth guardrails. IEEE service concentrated in AI readiness for critical infrastructure: Co-Chair of the IEEE PES Long Range Planning Committee Subcommittee 5 on Emerging Technologies, Digitalization, and AI Readiness; Vice Chair on the published IEEE-SA Industry Connections white paper IC25-004-01 on Grid Readiness for Data Center Deployment (January 2026); Section Chair and Editor on the forthcoming IC25-004-02 on Data Center Design for IT Load Protection. Two U.S. Patents Filed 2026 on AI Agent Safety; 22 issued U.S. patents across a 29-year career at Bell Labs and AT&T including five years as Director of Data Science and Artificial Intelligence.

AGENT ARCHITECTURE AND GOVERNANCE

Three production agentic AI platforms publicly verifiable at github.com/martymcenroe—A+ contributor grade, 10,000+ contributions in the last year, 64 repositories contributed to.

AssemblyZero—Multi-Agent Orchestration with Cross-Vendor Adversarial Verification

- Five-stage LangGraph StateGraph pipeline (Triage→Design→Spec→TDD→PR) coordinating 12+ concurrent Claude and Gemini agents with typed state, conditional routing, retry-with-backoff, and crash recovery via serialized checkpoints.
- Cross-vendor adversarial verification—Claude generates, Gemini reviews—catches hallucinations across model families.
- Five-layer evaluation framework: execution-based verification (pytest exit codes drive routing), AST structural analysis, cross-model review, stagnation detection, longitudinal learning from accumulated verdicts.
- 740 issues processed, 88% closure rate, 434 pull requests merged, 5,583 tests. Stack: Python, LangGraph 1.0, DynamoDB, GitHub Actions.

Hermes—Autonomous Production Agent for Recruiter Triage

- Production autonomous email agent for recruiter triage on Cloudflare Workers + AWS Bedrock. Architectural safety: deterministic first-contact template—zero AI on the highest-stakes message—audit queue gating AI-proposed actions, 10-state conversation state machine with escalation triggers (legal threat, message-count overflow, VIP allowlist), and operator-takeover flag (is_human_managed) for manual control.
- Two-stage Bedrock pipeline on returning conversations: Haiku for intent classification + entity extraction (10 intents), Sonnet for generation with full context injection. 60-120 second human-like send delay.
- Pipeline of quality gates on every output: anti-hallucination, sensitive-data pattern detection, AI-disclosure prevention, hostile-tone (27-phrase blocklist). Hybrid-cloud boundary at SQS—Cloudflare Worker ingests + pre-retrieves RAG, AWS Lambda processes—chosen for autonomous-agent reliability: at-least-once delivery, retry-on-failure, dead-letter quarantine. Form-fill gated behind verified GitHub star.
- Stack: Cloudflare Workers, AWS Lambda, SQS, D1, Vectorize, R2, Bedrock, SES. ~$5-7/month at hundreds of messages.

Aletheia—Consumer AI with Layered Guardrails

- Seven-stage handler pipeline for consumer-facing AI product with independent short-circuiting at each stage. Two-layer guardrail system: deterministic denylist followed by semantic LLM classifier with five-category taxonomy returning confidence scores.
- Tiered model routing: Nova Micro screens for layered-meaning potential, Claude Opus performs deeper multi-dimensional analysis on user demand.
- Privacy-constrained observability—operational metadata only, no content logging.
- Stack: AWS Lambda, Bedrock, DynamoDB, CloudFront, Chrome and Firefox extensions.

Maieutics—Methodology and Tooling for AI-Assisted Technical Authorship

- Methodology and tooling for AI-assisted technical authorship at institutional scale. The workflow constrains AI use to assistance and forces substantive human contribution at every step, so the resulting work is defensibly the author's under inspection.
- Three core components: per-document audit-trail specification, semantic source-verification that catches paraphrase crossing into copying, and a workflow checklist mapping each step to demonstrable human contribution.
- Legal anchor: USPTO Inventorship Guidance for AI-Assisted Inventions and the Pannu v. Iolab Corp. significant-contribution analysis.
- Status: private preview, active development. License: PolyForm Noncommercial 1.0.0.

sentinel-rfc—Agent-Context Permission Bit Architecture

- Active applied research program proposing agent-context permission bits as an architectural alternative to the "agent inherits user permissions" model that creates excessive blast radius for autonomous agents.
- Scope includes manifest specification, capability authorization design, and prototype work targeting filesystem-level enforcement.

AI & Data Science Architecture Review Board (AT&T)—Enterprise Governance Precedent

- Founding chair of the enterprise AI and Data Science Architecture Review Board, establishing governance gates for AI systems across business units of a Fortune 10 enterprise.
- Evaluated frameworks, vendors, and architectures for security, interoperability, licensing, and cost. Drove standardization of Python and R as enterprise-approved languages.
- Deployed and administered Stack Overflow Enterprise to 31,000 users, navigating Technical Architecture Board approvals, security reviews, and privacy compliance.

IEEE LEADERSHIP AND STANDARDS

Co-Chair, IEEE PES Long Range Planning Committee, Subcommittee 5 2026–present

Emerging Technologies, Digitalization, and AI Readiness. Reports to PES Governing Board via LRPC Chair (PES President-Elect). Vision Document due July 2026 PES General Meeting.

Vice Chair, IC25-004-01—Industry Standards for Grid Readiness for Data Center Deployment 2025–2026

IEEE-SA Industry Connections, Data Centers: Standards Needs Analysis and Recommendations. Published January 29, 2026 (ieeexplore.ieee.org/document/11366058). Recommendations triggered an IEEE SETCom study group (approved February 26, 2026) developing Project Authorization Requests for five new IEEE standards on data center interconnection, performance, modeling, co-located generation, and flexibility.

Section Chair and Editor, IC25-004-02 Section 5—Data Center Design for IT Load Protection 2025–2026

Section 5: Consumption Patterns and Disturbance Response. IEEE-SA Industry Connections; publishing 2026.

Section Author, IEEE PES PSCCC S17—Use of Software Bill of Materials (SBOM) in the Energy Sector 2025–present

Section: Use Cases in the EPS Industry. Six use-case analyses: procurement vs. maintenance vulnerability assessment, OSS licensing, country-of-origin and embargo risk, system-level SBOM, legacy devices, and SBOM scope beyond IEDs/RTUs. Citations span NIST SP 800-161 r1, NERC CIP-010/013, CISA/NSA/NTIA, NDAA §889, BIS Entity List, and peer-reviewed substation BoM literature.

PROFESSIONAL EXPERIENCE

THRIVETECH.AI – Plano, TX June 2022–present

Founder & President

Independent AI architecture practice and C2C consulting vehicle; holds the independent invention portfolio. Active products: **AssemblyZero**, **Hermes**, **Aletheia**, **Chiron** (AI governance study SaaS for IAPP AIGP preparation), and **Clio** (Chrome extension for extracting LLM conversations to structured JSON).

TEXAS DEPARTMENT OF TRANSPORTATION – Austin, TX September 2023–April 2024

AI Strategic Consultant

- Authored 76-page AI strategic plan encompassing 32 epics, 213 use cases, governance controls, risk assessments, and AI ethics framework for state-wide transportation agency.
- Conducted stakeholder interviews across every division and district; applied LDA topic modeling to surface themes and identify high-priority AI applications meeting federal and state regulatory requirements.
- Translated technical AI capabilities into business-value propositions for non-technical executives; established phased implementation roadmaps balancing feasibility, compliance, and organizational change.

AFINITI – Washington, DC March 2021–June 2022

Vice President, Product

Product and security leadership for AI-powered customer-agent pairing platform optimizing enterprise contact center operations through behavioral analytics and machine learning.

- Led security architecture initiatives on a ~1M-line, 10-year-old C/C++ codebase: SonarQube vulnerability remediation, penetration test remediation, architecture reviews, API security hardening.
- Defined cross-functional product development processes spanning engineering, AI research, MLOps, and go-to-market. Hands-on DevOps across GCP and Azure (Docker, Jira, CI/CD).
- Set product strategy and prioritization across quarterly and sprint planning cycles, balancing ML algorithm investment against operational improvements and production issues.

AT&T – Plano, TX September 1991–October 2020

Director, Data Science & Artificial Intelligence September 2017–October 2020

Hands-on architect and individual contributor leading enterprise-scale AI/ML solutions. Led 45-person cross-functional team, $9M annual labor budget, $10M+ recurring annual savings, $20M+ in initiative funding secured through SVP-level partnerships.

- **TAVI**—NLP customer-service chatbot (Python, TensorFlow, BERT, AWS Lambda, DynamoDB). 10,000+ daily interactions at 92% accuracy. Reduced costs $4.5M in 15 months; 30% first-call resolution improvement.
- **SNAP**—computer vision for field operations (Python, Azure CV, Core ML, Azure DevOps). First cloud-native AI solution at AT&T enterprise scale. $5.8M annual savings; 30% field productivity improvement.
- **EasyBake**—no-code chatbot platform on Azure with API and RPA overlays to dozens of AT&T enterprise systems. Three internal production customers. Pre-LLM conversational AI architecture.
- **Label Quest**—gamified human-in-the-loop labeling system used by dozens of internal data science teams, supporting text, image, and video labeling with real-time leaderboards, accuracy tracking, and consensus quality gates.
- Delivered 90+ proof-of-concept solutions over three years; security protocols, audits, and vulnerability remediation for all AI deployments.

Lead Member Technical Staff, Enterprise Architect May 2013–August 2017

- Founding chair of the AI & Data Science Architecture Review Board (described in Section above).
- Product manager for 'RCloud' internal data science platform; led training on Python, R, APIs, and AI architectures.
- Led tech evaluations setting enterprise direction: R vs SAS, MicroStrategy vs Tableau, GitHub vs Atlassian, AWS vs Azure vs GCP.

Additional Roles (1991–2013)

AT&T Bell Labs R&D (Holmdel, NJ; 1991–1995) • Director of Program Management, IP Backbone • Director of Emerging Video Services, Corporate Strategy • Network Operations & Engineering • Wireless communications development (multiple patents) • Resolved year-long migration crisis with 100-fold efficiency improvement in 45 days • Secured SVP funding for AI initiatives.

LICENSURE

Licensed Professional Engineer—State of Texas

CERTIFICATIONS

AWS—7 Certifications including:

- Machine Learning–Specialty
- Solutions Architect–Professional
- Data Analytics–Specialty
- Data Engineer Associate

Microsoft Azure:

- DevOps Engineer Expert
- AI Fundamentals
- Security, Compliance & Identity Fundamentals

CISSP & **CCSP**—(ISC)² | **PMP** & **PgMP**—PMI | **IEEE Senior Member** & **Wireless Communications Professional (WCP)**

EDUCATION

Master of Computer Science, University of Illinois Urbana-Champaign (2017)

Master of Science, Electrical Engineering, University of Maryland, College Park

Bachelor of Science, Electrical Engineering, University of Maryland, College Park

PROFESSIONAL MEMBERSHIPS

IEEE Senior Member • International Association of Privacy Professionals (IAPP) • (ISC)² • Project Management Institute (PMI) • Open Worldwide Application Security Project (OWASP)

PATENTS

Two U.S. Patents Filed 2026—AI Agent Safety. Independent inventor and applicant; active applied research program in autonomous-agent permission and capability architecture.

22 issued U.S. patents spanning AI, cloud computing, cybersecurity, IoT, and wireless communications, plus 9 pending applications. Domains include document analysis using ML and neural networks, customer service automation, sentiment detection, autonomous vehicle systems, edge network management, and biometric systems.